

# Software Test Plan (STP) – Hostel Management System

Project: Hostel Management System

Version: 1.0

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## 1. Introduction

**Purpose:** This document defines the Software Test Plan for the Hostel Management System (HMS) v1.0. It specifies the testing objectives, scope, strategy, test environment, schedule, deliverables, responsibilities, risks, suspension/resumption criteria, and requirements traceability approach.

**Scope:** Testing covers user authentication and role-based access, student hostel records, hostel/block/room management, room allocation, occupancy tracking, check-in/check-out, room transfer requests, complaint management, administrative approval workflows, occupancy/vacancy reporting, search and record retrieval, audit logging, performance, reliability, data integrity, usability, recovery, and security requirements.

**Out of scope:** University-wide academic administration and payment processing beyond possible future APIs, as explicitly excluded by the SRS.

**References:** Team\_5\_Hostel\_Management\_System\_SRS v1.0 (03-09-2026) and the supplied Software Test Plan template/sample.

**Definitions:** HMS (Hostel Management System), UI (User Interface), API (Application Programming Interface), DB (Database), RBAC (Role-Based Access Control), NFR (Non-Functional Requirement), RTM (Requirements Traceability Matrix), HTTPS, TLS, TC (Test Case), UAT (User Acceptance Testing).

## 2. Test Items

- Authentication and Access Control module
- Student and Room Management module
- Room Allocation and Stay Management module
- Room Transfer Request module
- Complaint Management module
- Reports, Monitoring and Audit module
- Performance and Reliability behavior
- Data Integrity and Recovery mechanisms
- Web UI on supported desktop and mobile browsers
- Security controls including authentication, RBAC, password storage, HTTPS/TLS, and input validation

### 3. Features to be Tested

Features mapped to the SRS requirement IDs:

| SRS Requirement(s) | Feature                                                                                | Primary Test Case(s)              |
|--------------------|----------------------------------------------------------------------------------------|-----------------------------------|
| HMS-F-001–003      | Authentication and role-based access                                                   | TC-Auth-01 to TC-Auth-03          |
| HMS-F-010–011      | Student, hostel, block and room records                                                | TC-STU-01, TC-RM-01               |
| HMS-F-020–025      | Room allocation, allocation status, capacity, duplicate allocation, check-in/check-out | TC-AL-01 to TC-AL-06              |
| HMS-F-030–032      | Room transfer request and approval workflow                                            | TC-TR-01 to TC-TR-03              |
| HMS-F-033–034      | Complaint submission and status/resolution updates                                     | TC-CMP-01 to TC-CMP-02            |
| HMS-F-040–042      | Occupancy/vacancy reports, pending items, audit logging                                | TC-RPT-01 to TC-RPT-02, TC-AUD-01 |
| HMS-NF-001         | Normal-load response time                                                              | TC-Perf-01                        |
| HMS-NF-002         | Monthly availability                                                                   | TC-REL-01                         |
| HMS-NF-003         | Data consistency and capacity integrity                                                | TC-DATA-01                        |
| HMS-NF-004         | Responsive and understandable UI                                                       | TC-UX-01                          |
| HMS-NF-005         | Backup and restoration                                                                 | TC-REC-01                         |
| HMS-SR-001–005     | Security controls                                                                      | TC-SEC-01 to TC-SEC-05            |

### 4. Features Not to be Tested

- University-wide academic administration, because it is outside the stated HMS scope.
- Payment processing beyond possible future APIs, because the SRS excludes it from the current system scope.
- Optional future ID-card or biometric device integration, because these are described as future/optional interfaces.
- Future external APIs and notification/email integrations unless implemented in the v1.0 build.
- Internal implementation details that are not externally observable requirements, unless needed to verify an SRS requirement.

## 5. Test Approach / Strategy

### 5.1 Test Levels

- Unit testing: Verify individual application components such as authentication, validation, allocation rules, complaint handling, reporting logic, and audit logging.

- Integration testing: Verify interactions among the web UI, application services, authentication/authorization components, and MySQL database.
- System testing: Verify complete end-to-end HMS workflows for Student, Warden, Hostel Administrator, and System Administrator roles.
- Acceptance testing (UAT): Verify the SRS acceptance criteria and the acceptance suites identified in Section 6 of the SRS.

## 5.2 Test Types

- Functional testing – verify all HMS functional requirements and business rules.
- Regression testing – repeat affected tests after changes to allocation, access control, records, reports, or related modules.
- Performance testing – verify the 90th-percentile response-time requirement under normal load.
- Reliability testing – verify the stated monthly availability target, excluding scheduled maintenance.
- Data integrity testing – verify allocation/occupancy consistency, uniqueness constraints, and room-capacity rules.
- Usability testing – verify supported desktop and mobile browser interfaces are responsive and understandable.
- Recovery testing – verify backup execution and restoration of system data.
- Security testing – verify authentication, RBAC, password hashing, HTTPS/TLS, and input validation/sanitization.

## 5.3 Entry Criteria

- A testable HMS build is available.
- Application and database environments are configured.
- Representative test data and required user roles/accounts are available.
- Planned test cases have been reviewed.
- No known environment issue prevents execution of the planned test suites.

## 5.4 Exit Criteria

- All planned test cases have been executed.
- All high-priority functional requirements are implemented and verified.
- No critical NFR or security failures remain.
- The RTM shows the planned requirements mapped to executed test cases and their status.
- Critical/blocking defects identified during the planned cycle are resolved or formally accepted for deferral.
- SRS acceptance criteria are satisfied for the release under test.

## 5.5 Security Validation

- TC-SEC-01: Verify protected student and administrative functions deny unauthenticated access.
- TC-SEC-02: Verify Student, Warden, and Administrator permissions and prevent unauthorized role access.
- TC-SEC-03: Inspect password storage and verify that plaintext passwords are not stored.
- TC-SEC-04: Verify production client-server communication uses HTTPS/TLS.
- TC-SEC-05: Submit malformed and injection-oriented input to applicable fields and verify validation/sanitization behavior.
- Verify important administrative actions generate auditable records containing user, action, and timestamp information.

## 6. Test Environment

The SRS specifies a web application running on modern browsers, an application server using Python (Django/Flask) or PHP (Laravel), a relational database such as MySQL, and a networked server environment.

| Environment Component | Planned Configuration                                                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Client                | Desktop and mobile devices with supported modern web browsers                                                                                                                                          |
| Application Server    | Python (Django/Flask) or PHP (Laravel), according to the implemented project stack                                                                                                                     |
| Database              | Relational database; SRS specifies MySQL as the software interface                                                                                                                                     |
| Network               | Networked client-server environment; HTTPS/TLS for production communication                                                                                                                            |
| Authentication        | Authentication and authorization components with protected sessions/tokens                                                                                                                             |
| Test Data             | Registered students, authorized staff, hostels, blocks, rooms, allocations, transfer requests, complaints, and audit events                                                                            |
| Proposed Test Tools   | Browser developer tools and/or Selenium for UI testing; Postman or equivalent for exposed APIs; JMeter or equivalent for performance testing; project defect tracker/spreadsheet for defect management |

Tool selection is proposed for this test plan; the SRS does not mandate particular testing tools.

## 7. Test Schedule

The following is a proposed schedule because the SRS provides requirements and acceptance criteria but does not specify testing dates.

| Milestone                      | Proposed Timing | Main Activities                                               |
|--------------------------------|-----------------|---------------------------------------------------------------|
| Test plan and test case design | Week 1          | Review SRS, derive test cases, prepare RTM                    |
| Test environment setup         | Week 2          | Configure application/database, browsers, roles and test data |

|                                                       |           |                                                 |
|-------------------------------------------------------|-----------|-------------------------------------------------|
| Unit and integration testing                          | Weeks 3–4 | Module and interface verification               |
| System / functional testing                           | Weeks 5–6 | Execute functional suites and regression tests  |
| Security, performance, usability and recovery testing | Week 7    | Execute NFR and security suites                 |
| Defect retest and regression                          | Week 8    | Verify fixes and rerun impacted tests           |
| UAT / acceptance testing                              | Week 9    | Run acceptance suites against SRS criteria      |
| Test closure                                          | Week 10   | Metrics, RTM update, final summary and sign-off |

## 8. Test Deliverables

- Software Test Plan (this document)
- Test Cases – functional, non-functional, security, and acceptance tests
- Test Scripts – automated/manual execution procedures where applicable
- Test Data
- Requirements Traceability Matrix (RTM)
- Test Execution Logs
- Defect Reports
- Regression Test Results
- Test Summary Report
- UAT / Acceptance Test Results

## 9. Roles and Responsibilities

| Role             | Name           | Responsibility                                                                                                                |
|------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------|
| Tester/Developer | R Agilesh      | Support user-role setup, access configuration, prepare and maintain the test plan, coordinate execution, monitor test status. |
| Tester/Developer | R G Nithik     | Support deployment environment , design and execute test cases, maintain test data, record evidence, and log defects.         |
| Tester/Developer | R Muralidharan | Support environment setup, analyze defects, implement fixes, and support defect triage with backup/recovery checks.           |
| Tester/Developer | R Pooja        | Provide domain validation and participate in acceptance testing of hostel workflows, security verification.                   |

## 10. Risks and Mitigation

| Risk                                                | Impact                                                             | Mitigation                                                                                                  |
|-----------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Delay in stable HMS build                           | Test execution may be delayed.                                     | Request incremental/smoke builds and prioritize high-priority requirement testing first.                    |
| Incorrect or incomplete test data                   | Functional and integrity tests may give misleading results.        | Prepare controlled datasets covering valid, duplicate, boundary, full-capacity, and invalid cases.          |
| Environment/database downtime                       | Execution and performance testing may be interrupted.              | Maintain reproducible environment configuration and backups; record downtime in execution logs.             |
| Role/permission misconfiguration                    | Security and RBAC tests may not be valid.                          | Create dedicated Student, Warden, and Administrator test accounts and verify permissions before testing.    |
| Changes to allocation logic affect multiple modules | Regression defects may occur in occupancy, transfers, and reports. | Run allocation, occupancy, transfer, reporting, and data-integrity regression suites after related changes. |
| Performance target not met                          | HMS-NF-001 acceptance criterion may fail.                          | Profile common operations, identify bottlenecks, optimize, and repeat controlled load tests.                |
| Backup restoration failure                          | HMS-NF-005 may not be satisfied.                                   | Perform scheduled backup verification and a controlled restoration test before acceptance.                  |
| Security defects in authentication/input handling   | Unauthorized access or malformed-input risks may remain.           | Execute the complete security suite and retest every security defect after remediation.                     |

## 11. Assumptions & Dependencies

- The implemented HMS will follow the functional and non-functional requirements in SRS v1.0.
- The application will use one of the technology alternatives specified by the SRS and a relational database such as MySQL.
- Supported modern desktop and mobile browsers will be available for testing.
- Required test accounts and role assignments will be provided before security and workflow testing.
- Representative student, hostel, block, room, allocation, transfer, complaint, and audit data will be available.
- HTTPS/TLS configuration will be available in the production-like environment when TC-SEC-04 is executed.
- Backup functionality and the configured backup schedule will be available for TC-REC-01.

- Future/optional interfaces are dependencies only if included in the implemented v1.0 build.

## 12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment or database is unavailable for a period that prevents meaningful execution.
- The build is sufficiently unstable that a substantial portion of the planned test cases cannot be executed.
- A critical defect blocks a core workflow such as authentication, room allocation, or database access.
- Required test data, accounts, or dependencies are unavailable and cannot be worked around.

Resume testing if:

- Blocking/critical defects have been resolved or an approved workaround is available.
- The test environment and database are stable.
- Required test data, roles, and dependencies are available.
- A smoke test confirms that the build is suitable for continued execution.

## 13. Test Case Management & Traceability

The RTM shall maintain a direct mapping between each SRS requirement and one or more test cases. The SRS defines requirement IDs, acceptance criteria, and test-case references; this test plan organizes them into an execution-oriented mapping.

| Requirement   | Verification Focus                                                  | Test Case(s)                   |
|---------------|---------------------------------------------------------------------|--------------------------------|
| HMS-F-001     | Valid credentials create authenticated session                      | TC-Auth-01                     |
| HMS-F-002     | Invalid credentials rejected; protected access denied               | TC-Auth-02                     |
| HMS-F-003     | Unauthorized role access blocked                                    | TC-Auth-03                     |
| HMS-F-010     | Student record saved; duplicate ID rejected                         | TC-STU-01                      |
| HMS-F-011     | Room record, capacity and availability maintained                   | TC-RM-01                       |
| HMS-F-020–023 | Allocation, date/status, capacity rule, duplicate active allocation | TC-AL-01–TC-AL-04              |
| HMS-F-024–025 | Check-in/check-out and occupancy update                             | TC-AL-05–TC-AL-06              |
| HMS-F-030–032 | Transfer request, decision, and consistent occupancy update         | TC-TR-01–TC-TR-03              |
| HMS-F-033–034 | Complaint creation, status and resolution remarks                   | TC-CMP-01–TC-CMP-02            |
| HMS-F-040–042 | Reports, pending items, and audit records                           | TC-RPT-01–TC-RPT-02, TC-AUD-01 |
| HMS-NF-001    | 90th percentile response time $\leq$ 3 seconds under normal load    | TC-Perf-01                     |
| HMS-NF-002    | Monthly availability $\geq$ 99%, excluding scheduled maintenance    | TC-REL-01                      |

|                |                                                                     |                     |
|----------------|---------------------------------------------------------------------|---------------------|
| HMS-NF-003     | Allocation/occupancy consistency and capacity integrity             | TC-DATA-01          |
| HMS-NF-004     | Responsive and understandable desktop/mobile UI                     | TC-UX-01            |
| HMS-NF-005     | Backup and restoration                                              | TC-REC-01           |
| HMS-SR-001–005 | Authentication, RBAC, password hashing, HTTPS/TLS, input validation | TC-SEC-01–TC-SEC-05 |

## 14. Test Metrics & Reporting

Metrics collected:

- Percentage of planned test cases executed.
- Percentage of test cases passed, failed, blocked, and not executed.
- Requirement coverage: percentage of SRS requirements mapped to executed tests.
- Defect count by severity and status.
- Defect aging and average time to resolution.
- Regression pass rate after defect fixes.
- Performance results, including 90th-percentile response time for TC-Perf-01.
- Availability measurements for the evaluated period.
- Security test pass/fail results.
- Backup and restoration verification results.

Reports:

- Daily or per-cycle test execution status.
- Defect status report.
- Updated Requirements Traceability Matrix.
- Performance, security, and recovery test results.
- Final Test Summary Report.

## 15. Approvals

| Role      | Name       |
|-----------|------------|
| Developer | R Agilesh  |
| Developer | R G Nithik |



|                    |                   |
|--------------------|-------------------|
| Developer          | R Muralidharan    |
| Developer          | R Pooja           |
| Course Coordinator | Prof. Asha Kumari |

## Appendix A – Planned Test Case Catalogue

| Test Case ID | Test Case                                                                 | Requirement(s)         |
|--------------|---------------------------------------------------------------------------|------------------------|
| TC-Auth-01   | Login with valid registered student/staff credentials                     | HMS-F-001              |
| TC-Auth-02   | Login with invalid credentials and attempt protected access               | HMS-F-002              |
| TC-Auth-03   | Attempt functions using an unauthorized role                              | HMS-F-003 / HMS-SR-002 |
| TC-STU-01    | Create/update student record and verify duplicate student ID rejection    | HMS-F-010              |
| TC-RM-01     | Create/update room record and verify capacity/availability                | HMS-F-011              |
| TC-AL-01     | Allocate an available room to an eligible student                         | HMS-F-020              |
| TC-AL-02     | Verify allocation date and status                                         | HMS-F-021              |
| TC-AL-03     | Attempt allocation when room is at configured capacity                    | HMS-F-022              |
| TC-AL-04     | Attempt a second active allocation for the same student                   | HMS-F-023              |
| TC-AL-05     | Record student check-in                                                   | HMS-F-024              |
| TC-AL-06     | Record check-out and verify occupancy update                              | HMS-F-025              |
| TC-TR-01     | Submit room transfer request with reason                                  | HMS-F-030              |
| TC-TR-02     | Approve/reject transfer as authorized staff                               | HMS-F-031              |
| TC-TR-03     | Complete approved transfer and verify old/new occupancy                   | HMS-F-032              |
| TC-CMP-01    | Submit hostel complaint and verify unique reference/status                | HMS-F-033              |
| TC-CMP-02    | Update complaint status and resolution remarks                            | HMS-F-034              |
| TC-RPT-01    | Generate occupancy/vacancy report and compare totals with allocation data | HMS-F-040              |
| TC-RPT-02    | View/filter pending transfers and unresolved complaints                   | HMS-F-041              |
| TC-AUD-01    | Verify audit record for important administrative action                   | HMS-F-042              |
| TC-Perf-01   | Measure common-operation response time under normal load                  | HMS-NF-001             |
| TC-REL-01    | Verify availability measurement against monthly target                    | HMS-NF-002             |
| TC-DATA-01   | Verify allocation/occupancy consistency and capacity integrity            | HMS-NF-003             |
| TC-UX-01     | Verify responsive and understandable desktop/mobile UI                    | HMS-NF-004             |

|           |                                                                           |            |
|-----------|---------------------------------------------------------------------------|------------|
| TC-REC-01 | Verify scheduled backup and successful restoration                        | HMS-NF-005 |
| TC-SEC-01 | Verify unauthenticated access is denied                                   | HMS-SR-001 |
| TC-SEC-02 | Verify RBAC restrictions                                                  | HMS-SR-002 |
| TC-SEC-03 | Verify passwords are securely hashed and not plaintext                    | HMS-SR-003 |
| TC-SEC-04 | Verify HTTPS/TLS protection                                               | HMS-SR-004 |
| TC-SEC-05 | Verify validation/sanitization against malformed/injection-oriented input | HMS-SR-005 |